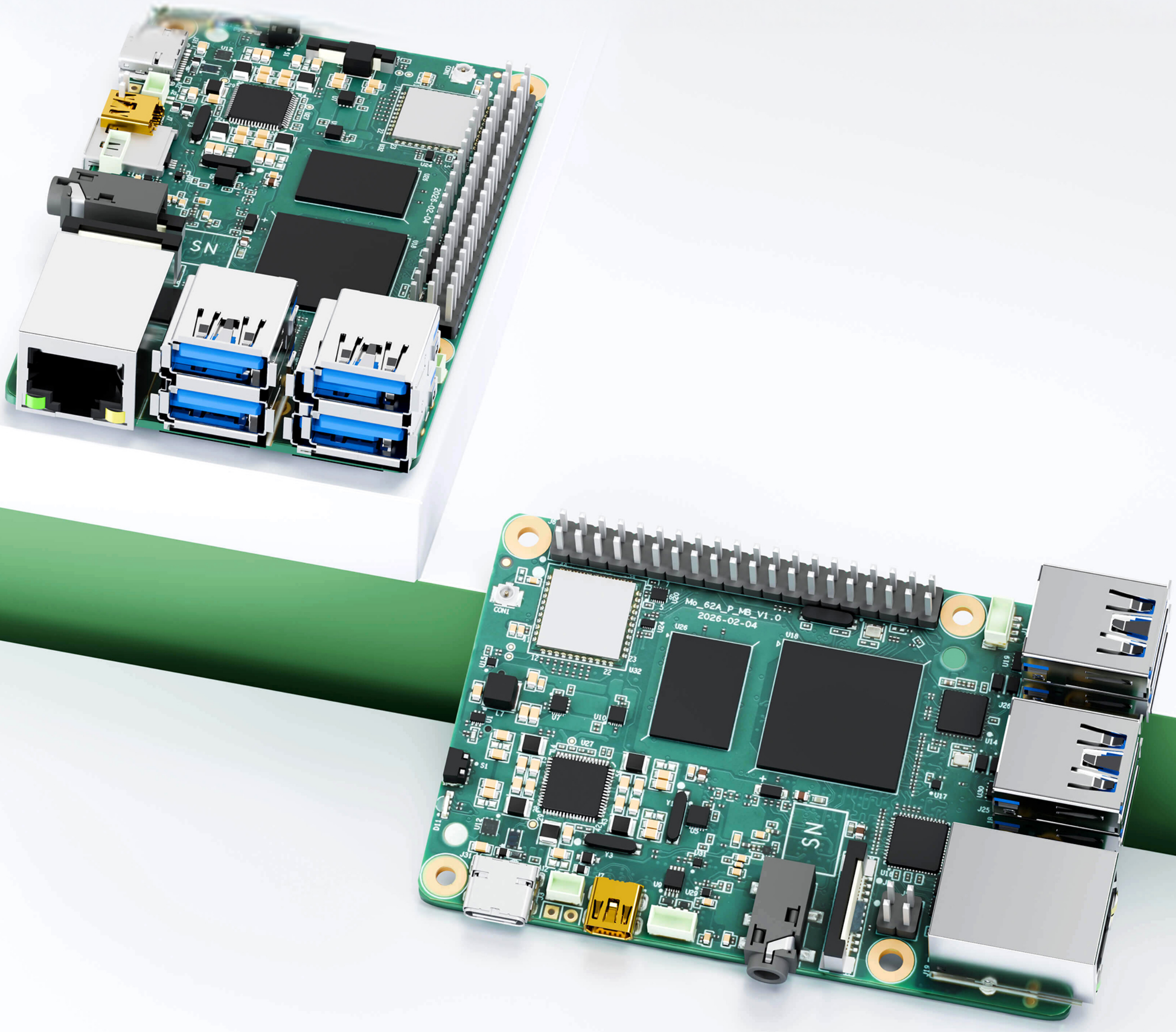
InHand Networks

Edge AI Computing, Open SDK, Debian Linux, HAT-Compatible



Mo 62A AI Single Board Computer

- Open SDK
- Linux Distribution
- HAT Expansion
- Edge AI

1. Product Overview

The Mo 62A is a 2 TOPS edge AI single-board computer based on the TI AM62A74 SoC, designed for AI vision boxes, intelligent cameras, and edge AI inference terminals.

Product Features:

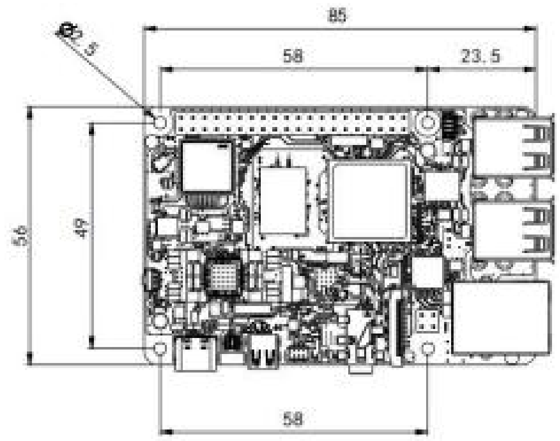
- **Edge AI Acceleration:** 2 TOPS on-device AI inference with C7x DSP and deep-learning accelerator
- **Open Ecosystem:** Debian Linux, open SDK, Python/C/C++ development with Docker support
- **Rich Interfaces:** Gigabit Ethernet, 4 × USB 2.0, micro HDMI, 4-lane MIPI CSI-2, 40-pin HAT connector
- **Wireless Connectivity:** Wi-Fi 5 dual-band and BLE for flexible deployment
- **HAT Compatible:** Standard SBC form factor, mechanically compatible with mainstream enclosures and accessories

Core Technical Specifications

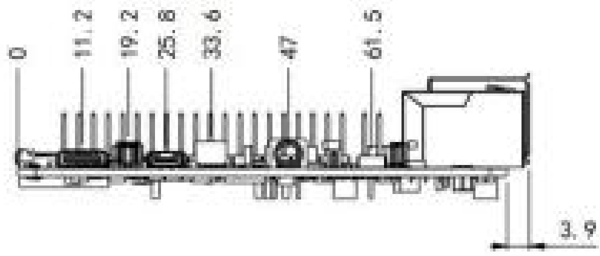
Technical Indicator	Specification
OS	Debian 13 (Embedded Linux)
AI Accelerator	C7x DSP + Deep Learning Accelerator, 2 TOPS
AI Runtime	TI TIDL, supports TFLite / ONNX
Wi-Fi	Wi-Fi 5 (802.11ac), dual-band 2.4 / 5 GHz
Bluetooth	BLE 4.2
Security	Secure Boot, TrustZone, OP-TEE, Hardware AES-256
CPU	4 × Cortex-A53 @ 1.4 GHz
RAM	LPDDR4 4 GB (default) / 2 GB / 8 GB

Technical Indicator	Specification
Interface	1 × GbE, 4 × USB 2.0, micro HDMI, MIPI CSI-2, 40-pin HAT
Power	USB Type-C 5 V / 5 A DC; ≤ 25 W
Dimensions (W × D × H)	85 × 56 mm (3.35 × 2.2 in)
Operating Temperature	0 °C ~ +50 °C (32 °F ~ +122 °F)

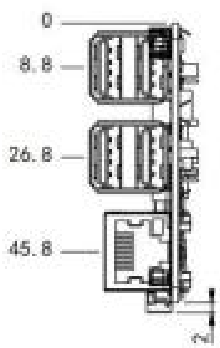
2. Product Dimensions



Front View



Side View



Interface View

Note:
1. All dimensions are in millimeters (mm), with inches (in) in parentheses.
2. All dimensions are approximate values for **reference only**.
3. The dimensions shown **shall not be used for production or processing**.
4. Dimensions shall comply with part and manufacturing tolerance requirements.
5. Dimensions are subject to change without notice.

3. Hardware Specifications

Category / Parameter	Specification
Hardware Platform	
CPU	TI AM62A74, 4 × Cortex-A53 @ 1.4 GHz
AI Accelerator	C7x DSP + Deep Learning Accelerator, 2 TOPS
ISP / Vision	On-chip ISP + VPAC (RGB-IR, WDR, LDC)
RAM	LPDDR4 4 GB (default) / 2 GB / 8 GB
Interface	
Ethernet	1 × Gigabit Ethernet
USB	4 × USB 2.0 Type-A
Display	1 × micro HDMI
Camera	1 × 4-lane MIPI CSI-2
Audio	3.5 mm jack + PCM via 40-pin connector
40-pin Connector	GPIO / I²C / SPI / UART / PCM, HAT-compatible
Fan Connector	1 × 4-pin fan connector
Button	1 × Reset button
Storage	microSD
Debug UART	1 × TTL UART
LED	PWR, USER
Wireless	
Wi-Fi	Wi-Fi 5 (802.11ac), dual-band 2.4 / 5 GHz
Bluetooth	BLE 4.2

Category / Parameter	Specification
Antenna	Wi-Fi / BLE: on-board snap-on antenna
Power	
Power Input	USB Type-C 5 V / 5 A DC
Power Consumption	25 W (MAX)
Mechanical	
Dimensions (W × D × H)	85 × 56 mm (3.35 × 2.2 in)
Weight	47 g (0.1 lb)
Housing	PCB
Cooling	Active fan (optional)
RTC	Support (battery backup)
Environmental	
Operating Temperature	0 °C ~ +50 °C (32 °F ~ +122 °F)
Storage Temperature	-20 °C ~ +70 °C (-4 °F ~ +158 °F)

4. Software Specifications

Category / Parameter	Specification
Operating System	
OS	Debian 13.2 Trixie
Kernel	Linux Kernel 6.12
AI & Vision	
AI Runtime	TI TIDL, supports TFLite / ONNX

Category / Parameter	Specification
Vision SDK	TI EdgeAI SDK
Camera Framework	V4L2
Display Framework	DRM / KMS
Network Features	
IP Application	TCP / UDP, ICMP, DNS, DHCP
IP Routing	Static routing
Security	
Secure Boot	Support
TrustZone	Support
OP-TEE	Support
Crypto Accelerator	Hardware AES-256
Development	
Languages	Python, C/C++
Libraries	OpenCV, GStreamer, NumPy
Package Manager	apt (Debian)
Open SDK	Supports custom system build by customer
System Management	
Remote Access	SSH
Firmware Upgrade	SD card flash
Debug	UART console

5. Ordering Information

Model Code Rule

Model code: Mo 62A-<Memory>

<Memory>: RAM Capacity

Product Models

Model	<Memory>: RAM Capacity
Mo 62A-2G	2 GB
Mo 62A-4G	4 GB
Mo 62A-8G	8 GB

6. Contact Us

- **Official Website:** [InHand Networks]
-